



**MANAV
RACHNA**

||vidyantarikshah||

MANAV RACHNA UNIVERSITY

(Department of Mechanical Engineering)



**L&T
EduTech**

Engineering the Future

PROJECT REPORT

ON

AGILE DEVELOPMENT MODEL (KANBAN BOARD)

(SMART WATER BOTTLE PROJECT)

SUBMITTED FOR

Critical Thinking, Design Thinking, Leadership and Teamwork



TASK NUMBER: 14



TOPIC: Agile Development Models

SUBMITTED BY:



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Course : B.Tech Mechanical Engineering



Semester : 2

SUBMITTED TO:



Faculty : L&T Specialists

INSTITUTION:



**Manav Rachna University
Faridabad, Haryana**

ACADEMIC YEAR:

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1. Introduction

Agile development is a modern approach to project management that focuses on flexibility, continuous improvement, and collaboration. Unlike traditional models such as the Waterfall model, Agile allows iterative development where tasks are completed in small, manageable stages.

In this project, the Agile methodology is implemented using a **Kanban Board** to design and manage the development of a **Smart Water Bottle**. The board visually represents tasks across different stages such as Backlog, To Do, In Progress, and Done, ensuring better workflow management.

Lucid Chart Link

https://lucid.app/lucidchart/956aec84-2466-4e54-ad0e-c05ca92a9bb6/edit?viewport_loc=-3593%2C-2280%2C6290%2C2571%2C0_0&invitationId=inv_96451c4d-ea8d-43a7-9122-6394ddde5fef

2. Objective of the Project

- To understand the Agile development methodology
- To apply Kanban principles in product development
- To design a structured workflow for a Smart Water Bottle
- To improve task tracking and project efficiency

3. Product Description: Smart Water Bottle

The Smart Water Bottle is designed to help users maintain proper hydration. It includes features such as:

- Water intake tracking
- Sensor-based monitoring (temperature, level)
- Mobile app integration
- Reminder notifications
- User-friendly and portable design

This product aims to solve the problem of irregular hydration among students and working professionals.

4. Agile Methodology (Kanban Approach)

The Kanban board divides the project into different stages:

4.1 Backlog

This stage includes all planned tasks and ideas:

- Design bottle shape
- Select sensors (temperature, water level)
- Plan mobile app features
- Decide materials
- Research competitors
- Battery selection
- Bluetooth module selection
- Marketing strategy planning

4.2 To Do

Tasks ready to be worked on:

- Create PCB design
- Design circuit diagram
- Create app UI wireframe
- Finalize sensor placement
- Create prototype sketch

4.3 In Progress

Tasks currently under development:

- Coding microcontroller
- Integrating sensors
- Developing mobile application

4.4 Done

Completed tasks:

- Basic concept finalized
- Initial research completed
- Project idea approved

5. Working of the Kanban Board

The Kanban board helps in:

- Visualizing workflow
- Managing task priorities
- Tracking progress in real time
- Identifying bottlenecks
- Improving team collaboration

Each task moves from **Backlog** → **To Do** → **In Progress** → **Done**, ensuring systematic development.

6. Advantages of Agile (Kanban Model)

- Flexible and adaptive to changes
- Continuous improvement
- Better time management
- Clear visualization of progress
- Efficient teamwork and coordination

7. Comparison with Waterfall Model

Aspect	Agile (Kanban)	Waterfall
Approach	Iterative	Linear
Flexibility	High	Low
Changes	Allowed anytime	Difficult
Feedback	Continuous	End-stage
Risk	Lower	Higher

8. Conclusion

The Agile Kanban approach proved to be highly effective in organizing and managing the Smart Water Bottle project. It allowed continuous tracking, flexibility, and efficient task execution. Compared to traditional methods, Agile ensures better productivity and adaptability, making it ideal for modern product development.

9. References

- L&T EduTech Course Material
- Agile Methodology Concepts
- Kanban Board Practices